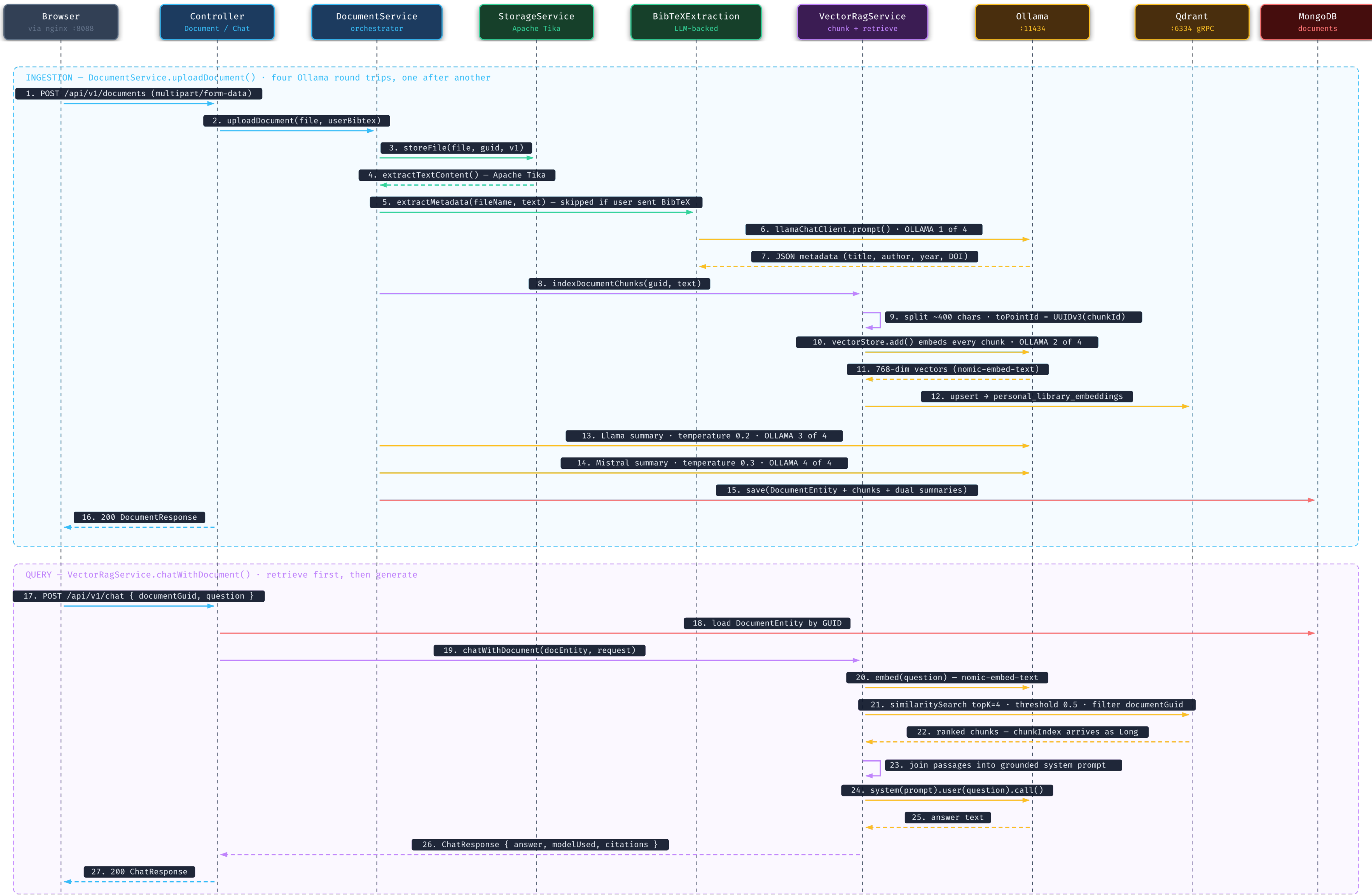


Personal Library – Backend RAG Sequence (Ollama + Qdrant)

Ingestion and query, both strictly sequential inside a single HTTP request – no concurrency anywhere in this path

UML Sequence



FALLBACK PATHS – all three fail soft

1. Qdrant indexing throws → logged as a warning only. The upload still returns 200 with nothing indexed.
2. Similarity search fails or returns nothing → falls back to docEntity.contentExcerpt as context.
3. Chat model unavailable → a fixed apology string, so the request never surfaces a 5xx to the browser.

TIMEOUT BUDGET – the sum, not one call

OLLAMA_READ_TIMEOUT · 10 min – applied PER CALL
nginx_proxy_read_timeout · 2160s (36 min)
LLM_TIMEOUT_MS (browser fetch) · 35 min

Upload chains up to FOUR Ollama calls, so the worst case approaches – and can exceed – the 35/36 min client and proxy budgets downstream.

CONFIGURED VALUES (read from source)

Chunking · paragraph split, ~400 chars, 50-char floor
Embeddings · nomic-embed-text, 768 dims, 8k context
Retrieval · topK 4, similarityThreshold 0.5
Models · app.models.llama=llama3.2 · mistral=mistral
Summaries · first 6,000 chars of the extracted text
Point IDs are UUIDv3 – Qdrant rejects suffixed GUIDs.